

# BRADEN FRUIN

<https://bradenfruin.github.io/>

## EDUCATION

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### BACHELOR OF SCIENCE, MECHANICAL ENGINEERING AND MATHEMATICS

Aug 2024 - May 2028

*University of Connecticut*

*Storrs, CT*

Cumulative GPA: 3.956/4.00

Credits: 57 (Junior Status)

Relevant Coursework: CAD, Multivariable Calculus, Elem Differential Equations, Linear Algebra, Applied Mechanics II, Intro to Computing Mech Engr, Applied Thermodynamics, Fluid Dynamics I

## SKILLS

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CAD, SolidWorks, Python, Excel, Time Management, Calculus, Problem Solving, Written Communication

## PROJECTS

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### S&P 500 Trading Algorithm

August 2025 – Present 2025

*Language: Python*

- Utilized historical price data and volume metrics to identify patterns and trends
- Implemented risk management strategies to minimize potential losses, including stop-loss orders and position sizing
- Back tested against historical data including historical constituents to ensure reliability

### UNIT CONVERTER

July 2025 -August 2025

*Language: Python*

- Unit converter that can handle conversions for Length, Weight and Temperature
- Learned about reusable functions, conditional Logic, Raising errors for unsupported units

### QR CODE GENERATOR

June 2025 – July 2025

*Language: Python*

- Convert URL to scannable QR code image
- Learned how to use QR code library, Manage Image files, Customize the Appearance of a QR Code
- Learned about Functions, Data Handling, Saving Files

## EXTRACURRICULAR

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### D.P. DOUGH

*Line Cook*

August 2024-Present

- 20-24 hours per week
- Developed and implemented efficient food prep schedules to ensure timely service during peak hours
- Worked to standardize recipes and prep procedures, ensuring consistent taste and presentation across all menu items.
- Demonstrated Excellent customer service during busy hours

### CLUB RUNNING

*August 2024-Present*

- Daily practices with meets/races on Saturdays
- 50-40 miles a week